

The Atlas 2: Arma Reforger

Terrain Creation Guide by Jakerod

Special Thanks

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The Atlas 2: Arma Reforger	1
Special Thanks	1
Introduction	2
Download & Launch Tools	3
Create New Project & Terrain	3
Basic Lighting Setup	5
Environment	5
Lighting	6
Fog & Haze	6
Post-Processing	6
Weather	6
Probe	6
Destruction	7
CameraManager	7
PreloadManager	7
ProjectileSoundsManager	7
AmbientSounds	7
RadioBroadcastManager	7
ForestSyncManager	7
SoundWorld	7
MusicManager	7
Main Viewport Camera Movement	7
Import Heightmap	8
Import Sat Map	8
Import Surface Mask	8
Painting a Surface Mask	9
Importing Roads	9
Sculpting Terrain	13
Create Rivers & Lakes	13

Create Roads	14
Create Forests	14
Basic Object Brush	14
Save Object Brush	15
Shoremap (WIP)	15
Map Labels	15
Sound Map	16
Ingame Map	16
Game Master (WIP)	17
Navmesh	17
Conflict (WIP)	18
Import Shapes with Forest Generators	19
Debug Menu	20
Export Shapes	21
Appendix A: New Engine Features	21
Appendix B: Tips and Tricks	21
Surface Painting	21
Satellite Image	21
Add multiple start locations to Game Master	21
Appendix C: Troubleshooting	22
Terrain is not at 0, 0, 0 and objects are already placed	22
Cannot hear any environmental sounds	22
Make sure you have a soundmap generated and an AmbientSounds prefab	22
Soldiers will not move or not move far	22
Can't move the camera past a certain point	22
Terrain has a big square over part of it	22
Vehicle can't move when previewing	22
Satellite image is pink	23
When the weather is set to Cloudy, the sky is black	23
After updating to 1.0, saved Object Brush no longer works	23
My Object Brush .conf is empty when opened in Workbench.	23
Grass is matching the color of the satellite map.	23
Objects Don't Load	24
Main Viewport is dark even on default terrains	24
Blocks of ground have weird color	24
Model underground is fully or partially lit despite the presence of a light probe	24
Appendix D: Everon	24

Introduction

New terrain tools for Arma Reforger were released at the same time as the game. The community has been picking it apart and I'm trying to gather some notes in one place. It's

entirely possible that I am picking bad options when importing things or that better options exist but I'm mostly guessing at what works at the moment or following what others did.

Since this Doc was created, there is some information that has been added to the [BI Studio Community Wiki](#) that should be helpful. It does not have everything this doc has at the time of writing this paragraph so both docs are useful.

Additional pages have been added to the community wiki here too: [World Editor Tools](#)

I have some old sample data from the Arma 3 Atlas Tutorial here that I'm using to test with: <https://pmc.editing.wiki/doku.php?id=arma3:terrain:the-atlas-by-jakerod>

Download & Launch Tools

1. Download Arma Reforger Tools on Steam (there is an option in Steam Library for tools near the top)
2. Make sure you launch Arma Reforger once first
3. Launch the tools (after closing the game)
4. In the Enfusion Workbench Launcher, press the Create New button and save your project somewhere outside of the Arma Reforger installation
5. Press Ok

Create New Project & Terrain

1. Go to Editors > World Editor
2. File > New World
3. Base scene
4. Press OK
5. In Resource Browser window - create "Worlds" folder in root of your project
6. Save the world (**don't save in a locked folder**. Ideally in the Worlds folder you just created)
7. In the Resource Browser, do a search for GenericTerrain_Default.et
8. Drag GenericTerrain_Default.et into the main viewport
 - a. `**TerrainEntity(TE)**`, class GenericTerrainEntity, is responsible for rendering terrain mesh and its textures
 - b. TE supports non-rectangular shapes of terrain as well, by clicking on "=" (equal sign) user may set different grid size for X and Z axis
9. Check the lower right of the window and make sure Object Properties is selected
10. Find the transformation section of the Object properties and set it to X: 0 Y: 0 Z: 0
 - a. Note: I recommend you make sure you do this but if you forget, you can move it later. Make sure any objects you have are grouped under the "terrain" layer and then you can change these numbers and everything should move with the terrain.
11. At the top of the Object Properties window, right-click on GenericTerrainEntity and select Create new terrain

12. Pick the name of your terrain
 - a. This is how the folder holding terrain data will be named. Example: if project is stored in world\Test.ent, then there will be world\Test\Name_of_Terrain folder
13. Pick your terrain grid size
 - a. User is allowed to set size from 32 up to 16384(hardware limit), it can be any even-number divisible by 32
 - b. Terrain grid size is count of terrain faces in rows/columns, remember that 512 faces is 513 vertices, this necessary to keep in mind when importing heightmaps.
 - c. Terrain grid size multiplied by Grid cell size determines the final size of terrain, final terrain size together with terrain's area can be found in the bottom part of window in the info section
 - d. Like any other entity, Terrain entity must always be placed completely within world's bounds(world entity). In case any part of terrain entity is sticking out of bounds rendering glitches may appear(eg. missing terrain blocks).
14. Leave block at default unless you have a good understanding of the following
 - a. Block is the smallest part of TE which gets LODed, in opposite to RV engine where terrain LODing was done on tiles level.
 - b. 1 block = 1 draw call
 - c. Even though surface mask(for rendering detail surfaces) is split and mapped per a tile, limit of maximum 5 detail surfaces is being applied to blocks
 - d. Enfusion engine currently supports 5 LODs per block
 - e. Each block consists of 32x32 faces(33x33 vertices) down to 2x2 faces(3x3 vertices)
 - f. Determines how many blocks per tile as well as tiles per terrain will be created
 - g. Total count of tiles(per terrain), tile size in kilometers(tile size) and tile area(tile area) information can be found in the bottom part of window in the info section
15. Grid cell size in meters (this value multiplied by terrain grid gives you the size of your terrain)
 - a. Distance between two neighboring terrain vertices
 - b. Lower the value, higher the detail of terrain
 - c. Increasing/decreasing this value affects the final size of terrain
16. Height scale - leave at default value
 - a. Explanation of why this should stay at default values from Branzik:
 - i. The terrain heightmap is internally in 16bits integer format and the height scale value is the difference between minimum delta, so e.g. difference between 100 and 101 in internal representation is 0.0315 as default. This value shows the maximum possible height of terrain , 0.03125×65535 (the 16 bit value) = ~2048 meters -> it's visible also in log box in "available height" message.
 - ii. Changing this value user affects elevation(vertical) range(see available height in the info section)
 - iii. Lower the value, higher the precision, but smaller vertical range(and vice versa)

17. Zero height to entity coord - leave at default value
 - a. How much of elevation range will be used by terrain under sea level(again, see available height in the info section)
18. Press Create
19. Make sure Terrain in the Hierarchy is selected
20. Go to the Object Properties tab
21. Scroll down to the Unsorted section
22. In the list of items with checkboxes next to them, locate where it says "Terrain" and check the box next to it
23. Save

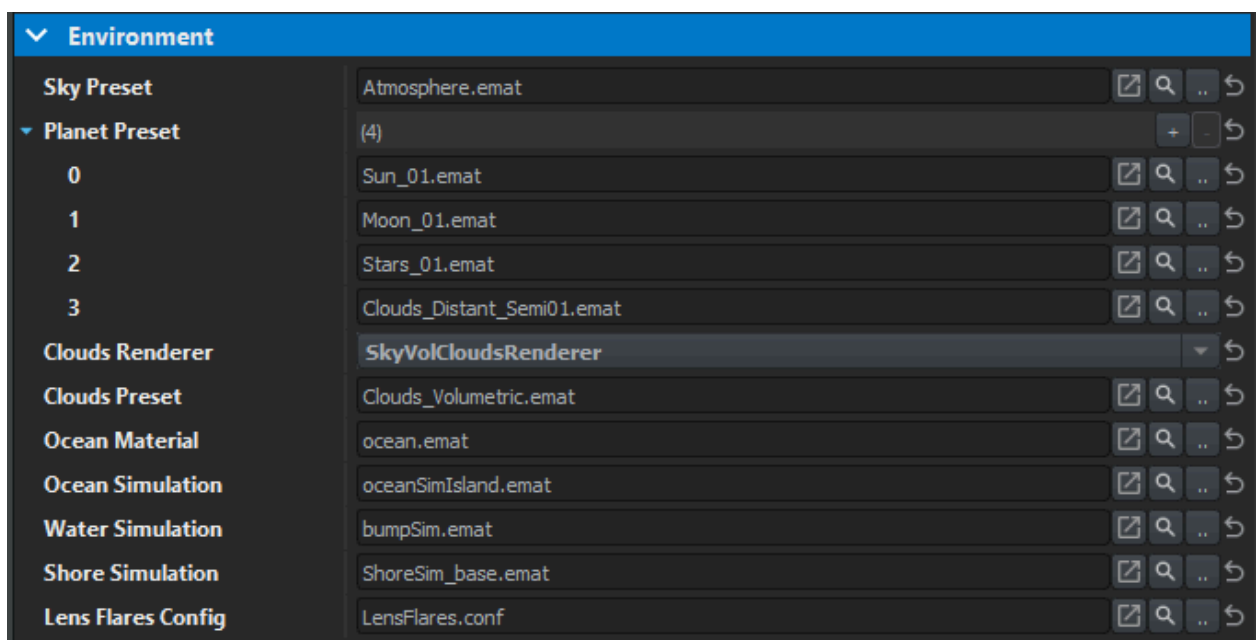
IMPORTANT: Do NOT rename the layer named "default" to anything else. Renaming it can break several functions.

Basic Lighting Setup

To avoid any crashes or loss of work, do the following lighting setup in the order that it is written

Environment

1. Save your work
2. Open your GenericWorldEntity
3. In the Environment tab in Object Properties, set up your configs to match these by selecting ArmaReforger in the Resource Browser, searching for the names in the text boxes below and then dragging them into the text boxes:



Safe practice is to to them from top to bottom to avoid engine crashes

Note: there are other ocean materials and ocean simulations that work too

Lighting

1. Save your work
2. In the Resource Browser, search for `lighting_default` and drag it onto your viewport
3. Inside its Object Properties, the only values that should be changed are it's Transformation values
 - a. Angle X - light source pitch angle
 - b. Angle Y - light source yaw angle
 - i. 0 - from South
 - ii. 90 - from West
 - iii. 180 - from North
 - iv. 270 - from East
 - c. Angle Z - does nothing

Fog & Haze

1. Save your work
2. Go into your Resource Browser
3. In the Resource Browser Search field, type in `foghaze_default.et`
4. Drag `FogHaze_Default.et` onto your viewport

Post-Processing

1. Save your work
2. Go into your Resource Browser and Navigate to ArmaReforger
3. In the Resource Browser Search field, type in `genericWorldPP_Default`
4. Drag `GenericWorldPP_Default.et` onto your viewport

Weather

1. Save your work
2. Go into your Resource Browser and type in `weather`
3. Navigate to ArmaReforger > Prefabs > World > Game
4. Drag `TimeAndWeatherManager.et` onto your viewport
5. Leave this alone, it is a fully functioning prefab made by Bohemia, and it includes all of the details needed for gamemodes

Probe

1. Save your work
2. Go into your Resource Browser and type in "envprobe"
3. Drag the `EnvProbe_Default.et` onto your viewport
4. Make sure the Probe entity is on land and in an area that best represents your overall terrain. If you put it over water everything turns blue.

Destruction

1. Save your work
2. Go into your Resource Browser and type in destructionManager
3. Drag `DestructionManager.et` into your world
4. Drag `MPDestructionManager.et` into your world

CameraManager

1. Save your work
2. Go into your Resource Browser and type in cameramanager
3. Drag `SCR_CameraManager.et` into your world

PreloadManager

- Add `PreloadManager.et`

AmbientSounds

- Add `AmbientSounds_Everon.et` or `AmbientSounds_Arland.et`

RadioBroadcastManager

- Add `RadioBroadcastManager.et`

ForestSyncManager

- Add `ForestSyncManager.et`

SoundWorld

- Add `SoundWorld_Base.et`
- See the [Sound Map](#) section to set this up

MusicManager

- Add `MusicManager_Base.et` (Do not add this until you make a Game Master scenario)

Main Viewport Camera Movement

- Rotate Camera - Hold RMB and drag
- Move Forward - Hold RMB + W
- Move Backward - Hold RMB + S
- Slide Right - Hold RMB + D or Hold MMB + Drag Left
- Slide Left - Hold RMB + A or Hold MMB + Drag Right
- Up - Hold RMB + Q or Hold MMB + Drag Back
- Down - Hold RMB + Z or Hold MMB + Drag Forward
- Change Camera Speed - Hold RMB - Use Scroll Wheel

- Orbit Position - ALT + LMB or ALT + RMB (if object is selected)
- Pan - Hold MMB and drag
- Zoom - Scroll wheel forward (in) or back (out)
- Note: There is a camera tab above the main viewport that has some useful items in it
 - Most people seem to like to change the Far Plane to a higher number so they can see further

Import Heightmap

1. Save your work
2. Press the Mountain button (Terrain Tool) at the top toolbar
3. Go to the Terrain Tool tab at the bottom of the right side of the window
4. Import Heightmap (should be asc)
5. Pick file
6. Uncheck invert z axis
 - a. Note: You may need to use this. I have found that my Arma terrains come in backwards if I don't but I have heard others say it isn't necessary. Double check your heightmap after import and make sure the height features are in the right places. Easiest way I have found so far to determine north, is set the camera to 0,0. I think that will be the south-west corner.
7. Uncheck resample heights
8. Import
9. Generate the Normal Map
10. Save

Import Sat Map

1. Save your work
2. Import Satellite Map
3. Select File (must be png)
4. Turn off linear color space (Note: if your sat map is too dark after importing with this turned off, try turning it on)
5. Press OK
6. Save
7. Reopen the project

Import Surface Mask

1. Save your work
2. Select the Terrain Tool from the toolbar
3. Go to the Paint tab/panel
4. Go to Resource Browser and navigate to ArmaReforger > Terrains > Common > Surfaces

5. Drag any surfaces you're interesting in using under the default material that is in the Paint panel on the right
 - a. Recommendation: add dirt type surfaces first and then grass type surfaces afterwards. This ordering matters to the parallax map. For the explanation on why, see the [Arma Reforger Modding Boot Camp #11 - Environmental Design video at timestamp 25:05](#)
6. You need a black and white surface mask where white determines where the surface should be
7. Right-click a surface, select priority surface mask import
8. Navigate to your surface mask in the new window
9. Press Open
10. The surface should now be applied
11. Save your work

Painting a Surface Mask

1. Save your work
2. Go to the Terrain Tool
3. Go to the Paint tab/panel
4. Check the "Show diags for block under cursor" option. This will show you which surfaces are in the surface block your cursor is currently over. The list of surfaces will appear in the upper left of the window. Note that you can only have **5 surfaces per block**. The Info & Diags tab next to Paint can be used to give a deeper breakdown of surfaces per block and can also be used to swap out surfaces for one another.
5. If you do not have any surfaces in the bottom of this panel or if there is only one named default, go to Resource Browser and navigate to ArmaReforger > Terrains > Common > Surfaces and drag in any surfaces you are interested in using.
6. Select a surface in your Paint panel by left-clicking on it
7. Go to the Main Viewport and left-click to paint
8. Note that you have some options in the upper part of the Paint panel for Strength, Radius, Falloff, Angle, Type of brush, and Shape of brush.
 - a. CTRL + Move Mouse will change the radius of the brush.
 - b. SHIFT + Move Mouse will change the strength of the brush.
 - c. CTRL + SHIFT + Move Mouse will change the Falloff of the brush.
 - d. CTRL + SHIFT + Mousewheel will change the Angle of the brush

Importing Roads (Optional)

1. Save your work
2. In order for this method to work, each shape in your shapefile requires an attribute that defines which type of forest it is. The only setup I have tested so far is the attribute of ID with each value being a number. It's possible other values will work but this tutorial is going to use that setup. I recommend using QGIS to add the ID attribute and a value.
3. Go to your project directory and create a folder called `roads` inside of it

4. Save or copy your .shp (along with .shx, .prj, and .dbf) file into the `yourProject\roads` folder
5. Go to the **Import Shapefile** button in the top toolbar
6. The SHP Path field is a path to your shapefile. Press the “..” button.
7. In the Choose m_sHPPPath (shp files) window, select your shapefile
8. Press OK
9. If your shapefile was made for Arma 3 or from GTT, you will want to add an X offset value of -200000.
10. If your shapefile is from a different source, for example OSM data you downloaded, you will need to find the offset to use. If for example your shapefile is a square that covers your whole terrain and the coordinate of the bottom left vertex s 251, 267 then your offset would be X: -251 and Z -267
11. In the ID Column Name field, type ID
12. Go to Resource Browser and search for “SCR_SHP”. There should be one file called `SCR_SHPPrefabDataList_ForestGenerator.conf`
13. Right-click on the file and go to **Duplicate to <yourProjectName>**
14. Select a name for this file. I normally do `SCR_SHP_RoadGenerator`
15. Press OK
16. The file should now appear in your project folder instead of a folder called `Configs\ForestGenerator`
17. In Resource Browser, right-click and select Show in Explorer
18. Copy the `.conf` and `.config.meta` file to your `roads` folder
19. Open the `.conf` file in Notepad or Notepad++ and copy and paste the following into the file:

```

SCR_SHPPrefabDataList {
  m_aPrefabsData {
    SCR_SHPPrefabData "{60E126E4108BE826}" {
      m_iID 1
      m_sPrefab
"{02AF8C5A31EC3A53}PrefabLibrary/Generators/Roads/Asphalt/RG_Road_A
sphalt_E_01.et"
    }
    SCR_SHPPrefabData "{60E126E562041A75}" {
      m_iID 2
      m_sPrefab
"{5E336AEB0923963F}PrefabLibrary/Generators/Roads/Asphalt/RG_Road_A
sphalt_E_01_DashedLine.et"
    }
    SCR_SHPPrefabData "{60E126E5796052E6}" {
      m_iID 3
      m_sPrefab
"{31086BE1AF790FC5}PrefabLibrary/Generators/Roads/Asphalt/RG_Road_A
sphalt_E_01_Narrow.et"
    }
    SCR_SHPPrefabData "{60E126E575DAFB46}" {
      m_iID 4
      m_sPrefab
"{6EFBB8BA8D28B57D}PrefabLibrary/Generators/Roads/Asphalt/RG_Road_A
sphalt_E_02.et"
    }
    SCR_SHPPrefabData "{60E126E57639010E}" {
      m_iID 5
      m_sPrefab
"{8B67F44381CD2216}PrefabLibrary/Generators/Roads/Asphalt/RG_Road_A
sphalt_E_03.et"
    }
    SCR_SHPPrefabData "{60E126E68C889110}" {
      m_iID 6
      m_sPrefab
"{ABC15429070F8391}PrefabLibrary/Generators/Roads/Cobblestone/RG_Ro
ad_Cobblestone_01.et"
    }
    SCR_SHPPrefabData "{60E126E68BB62198}" {
      m_iID 7
      m_sPrefab
"{2D9AEF10F5FEA98B}PrefabLibrary/Generators/Roads/Dirt/RG_Road_Dirt
_01.et"
    }
    SCR_SHPPrefabData "{60E126E681DB40AF}" {
      m_iID 8
      m_sPrefab
"{41CEDBF0493A26A5}PrefabLibrary/Generators/Roads/Dirt/RG_Road_Dirt

```

```

_02.et"
}
SCR_SHPPrefabData "{60E126E69FE4278D}" {
  m_iID 9
  m_sPrefab
"{6C770E9E60A49C05}PrefabLibrary/Generators/Roads/Dirt/RG_Road_Fore
st_01.et"
}
SCR_SHPPrefabData "{60E126E69BAB0FB2}" {
  m_iID 10
  m_sPrefab
"{47E7068FE332800D}PrefabLibrary/Generators/Roads/Dirt/RG_TrailDirt
_01.et"
}
SCR_SHPPrefabData "{60E126E69425F97B}" {
  m_iID 11
  m_sPrefab
"{98DF2267CDD17758}PrefabLibrary/Generators/Roads/Dirt/RG_TrailGrav
el_01.et"
}
}
}

```

3. Save the file above
4. In Resource Browser, navigate to your `roads` folder and confirm that the new file is shown there
5. In Resource Browser, right-click on your new `.conf` file (ex. `SCR_SHP_RoadGenerator.conf`)
6. Go to **Open in Resource Manager**. This will show the contents of the file.
7. You can left-click on the blue > signs and it will open each class. Each class has an ID and a Prefab. Any shapefile that has an attribute ID of 1 will get the `RG_Road_Asphalt_E_01.et` prefab assigned to it. An ID of 8 will assign `RG_Road_Dirt_02.et`.
8. These IDs and Prefab are editable.
9. If you wish, you can add a new entry by going to the top line where it says (11) and clicking the + button.
10. A new entry will be added to the bottom. You can open it and assign an ID and prefab.
11. If you made changes, go up to Workbench and select Save
12. Go back to the World Editor
13. From the resource browser, drag your `.conf` file into the Import Shapefile tool and into the Prefab Data List field where it says "set class"
14. Check the **Create As Spline** box
15. Select **Import SHP** and your shapes will be imported with road prefabs assigned to them

16. If your roads do not appear over your terrain, you may need to use the offset fields to align them.
17. Save the work

Sculpting Terrain

1. Save your work
2. In the Terrain Tool tab, go to the Sculpt tab at the top
3. Select the Sculpt option if it is not already selected
4. Up the strength if you want (especially if it is at 0)
5. Left-Click and drag on the terrain to raise
6. Left-ALT + Left-Click and drag on the terrain to lower
7. Holding Left-CTRL and dragging changes brush radius
8. Holding Left-SHIFT and dragging changes brush strength (or elevation on the Flatten tool)
9. Holding Left-SHIFT + Left-CTRL and dragging changes Falloff
10. Pressing 1 in the number row selected Sculpt
11. Pressing 2 in the number row selects Flatten
12. Pressing 3 in the number row selects Smooth
13. Pressing 4 in the number row selects Noise

Create Rivers & Lakes

1. Save your work
2. Select the Vector Entity Tool from the toolbar (looks like a squiggly line)
3. Go to the right window panel, look at the bottom, and select Vector Tool
4. Check the Snap to terrain box
5. Go to New Polyline
6. Left click onto your terrain where you want a river to be
7. A line segment will be generated
8. You can click on a vertex to select it and use the 3 arrows or 3 cube sides to move it around
9. You can add more vertices after the 2nd by CTRL+LMB to add a point after the selected point or ALT+LMB to add one before
10. Once you're done drawing, go to the Object Properties tab in the lower right panel
11. RMB on the PolylineShapeEntity entry (with the S next to it)
12. Go to Create River Entity in the new menu
13. There is a small chance you will need to go to the Resource browser and go to ArmaReforger > PrefabLibrary > Generators > Water > River and then drag and drop the `R_RiverMedium_01.et` onto your PolylineShapeEntity in the Hierarchy panel
14. If it's not working, check your Hierarchy window and see if moving the PolylineShapeEntity is subordinate to the Terrain. If so, try to move it so it is at the same level as the terrain which should be subordinate to world (to the best of my understanding)

15. NOTE: Process is similar for lakes. Just use the lake water instead of the river water and make sure that your polyline end is close to the polyline beginning. You can also use the Is Closed option in the Object Properties to view the polyline as if it is closed.

Create Roads

1. Save your work
2. Select the Vector Entity Tool from the toolbar (looks like a squiggly line)
3. Check the Snap to terrain box
4. Select New Spline
5. Click onto the terrain to start the road
6. Go to the Resource Browser
7. Navigate to ArmaReforger > PrefabLibrary > Generators > Roads
8. Pick the road type that you want and drag one of the options onto the SplineShapeEntity in the Hierarchy panel
9. Select the current line by clicking on a vertex and either move the two vertices you have using the arrow or cube movement widgets, or use CTRL+LMB or ALT+LMB (see top of Vector Tool pane) to add new vertices
10. You can change the width by selecting the SplineShapeEntity's Appearance/Material in the Hierarchy and going to the Object Properties panel on the right and finding Road Width in the Road section of the panel

Create Forests

1. Save your work
2. Draw or import polygons (see the Rivers and Lakes section for how to draw)
3. In the Resource Browser, navigate to ArmaReforger > PrefabLibrary > Generators > Forests
4. Drag the Forest Generator onto the Polyline in the Hierarchy list
5. It should populate the inside of the polyline with trees

Basic Object Brush

1. Save your work
2. Find the the Object Brush in your top toolbar
3. Open the Object Brush tab on the bottom right
4. Click on the set class bar next to Objects Config
5. After the Config opens to show an Object Array under it, then click the + on the right side
6. Give the Object Array a name (you will have to choose a name for each object you want to use)
7. The only settings here that really should be changed for a basic brush are:
 - a. Bot Distance - The radius around each object to allow for another object to spawn

- b. Random Pitch Angle - A random rotation angle between 0 and 180 degrees, this is useful for nature brushes
 - c. Align to Normal - The objects will be oriented off the surface they are painted on, instead of always staying upright
- 8. After you've created your Object Config, you can then tell the brush either avoid covering Objects, such as props or buildings, Roads, Rivers, and Powerlines

Save Object Brush

1. Save your work
2. Inside of the Resource Browser, go to your project folder and right-click
3. Select New Folder
4. In the text field type Brushes
5. Press OK
6. Double-left-click the Brushes folder to open it
7. Right-click and select Config File
8. In the text field of the Choose Name window, type in a name that describes the brush (ex. Low pine trees)
9. Press OK
10. Choose the class `SCR_ObjectBrushArrayConfig`
11. Drag and drop the new config into the text field next to the Object Brush's "Brush Config Path"
12. Press Save at the bottom of the Object Brush
13. Double-click on the `.conf` file you just saved to to open it in workbench and verify that your changes were saved
14. You will need to close out of Workbench in order to load the saved file whenever you make a change to it.

Shoremap (WIP)

Map Labels

This section explains how to add a map label that shows up in the 3D World of World Editor, Game Master, and on the 2D Map. It can be used to place town names, local names, water feature names, or many other types of map symbols.

1. Save your work
2. Go to the Create tab
3. Locate a GenericEntity and drag it onto the position you want the map label to appear
4. Go to Object Properties
5. Go to the field where it says (no name) next to GenericEntity
6. Type the name of your location (ex. Quarry or Fake_Moon_Landing_Site)
7. Right-click on the GenericEntity in the Object Properties
8. Go to Add Component

9. Find SCR_EditableCommentComponent
10. Right-click on the GenericEntity in the Object Properties again
11. Go to Add Component
12. Find SCR_MapDescriptorComponent
13. Select the SCR_EditableCommentComponent
14. Under the Editable Entity label, Set Entity Type to COMMENT
15. Set Auto Register to ALWAYS
16. Click the arrow next to Flags
17. Check the boxes next to LOCAL, NON INTERACTIVE, SLOT, NON DELETABLE, and IGNORE LAYERS
18. Under Visualization, next to UI Info, select set class and set it to SCR_EditableCommentUIInfo
19. Under Visualization, set the Name to the desired display name (ex. Quarry)
20. Select SCR_MapDescriptorComponent
21. Set Display Name to the desired display name (ex. Quarry)
22. Set Main Type to your desired label type (ex. Name Local)

Sound Map

1. Save your work
2. If you do not have one of them added already, add `AmbientSounds_Everon.et` or `AmbientSounds_Arland.et`
3. If you do not have it added already, add `SoundWorld_Base.et`
4. Select the Export Map data tool
5. Click the .. button next to Destination Path
6. Select your Terrain folder
7. Next to Type, select Soundmap
8. Press Export
9. Select the `SoundWorld_Base` entity
10. In Object Properties, go to where it says Soundmap and press the .. button next to it
11. Locate and select your soundmap (`terrainName.smap`)
12. You might need to restart the entire workbench to start hearing it

Ingame Map

This section is based on the information found on the [Community BIKI](#).

1. Save your work
2. Go to the Resource Browser
3. In the Resource Browser Search Field, type `MapEntity_Default.et`
4. Drag the `MapEntity_Default.et` into the main viewport
5. Save your project
6. Close out of Workbench
7. Reopen your project
8. Go to the Export Map Data tool in the toolbar

9. Locate the Export Map Data tab if necessary
10. Press the .. button next to Destination Path and set it to your terrain folder
11. Set the Type to Geometry 2D
12. Press Export
13. Switch the Type option to Rasterization
14. Press Export
15. Locate the .tga that was generated in your project directory and convert it to .png using your favorite image editing program (GIMP, Photoshop, TexView, etc.)
16. Go to Resource Browser and locate your .png file in your project directory
17. Right-click on the .png file in resource browser and select Register and Import
18. Wait for a .edds file to appear
19. Locate your MapEntity and select it
20. Go to Object Properties
21. Point the Map Geometry Data field to your .topo file using the .. button
22. Point the Satellite background image to your .png file using the .. button
23. NOTE: You have to test the map in a Game Master subscene. If you try to preview it in the main scene it doesn't work for some reason.

Game Master (WIP)

1. Save your work
2. In your project directory folder structure, create a folder called GameMaster inside your Worlds folder
3. Select New in the upper left of the window
4. In the Create World window, select Sub-scene and press OK
5. Go to File > Save World as
6. Save your file as GM_<terrainName> inside of the GameMaster folder
7. Go up to Plugins along the top toolbar and select Game Mode Setup
8. Use the `GameMaster.conf` template
9. Press Next
10. Press Scan world
11. Press Next
12. Press Create entities
13. Press Next
14. Press Create header
15. Press Next
16. Press Close
17. Setup your new MapEntity by following the steps in the Ingame Map section above
18. Setup your SCR_AIWorld entity by following the Navmesh section below
19. **Mission Header file (WIP)**

Navmesh

This should be done in your Game Master Subscene and assumes you already have an SCR_AIWorld entity from using the Game Mode Setup plugin.

1. Save your work
2. Select the Navmesh tool
3. In the Navmesh tool window, in the dropdown next to Navmesh, select Soldiers
4. Press Generate
5. Make sure Soldiers is the selected Local navmesh and then press OK in the Connect window
6. Press Generate
7. Press OK
8. Press OK
9. Wait for the Navmesh to generate
10. Press the Save Navmesh button
11. Press Other
12. Save it into your project's NavMesh folder with the name terrainName_soldiers (if you don't have a NavMesh folder create one)
13. Go back to the Navmesh tool
14. Next to Navmesh, select LowRes from the dropdown
15. Press Generate
16. Select LowRes from the drop down if it is not selected already
17. Press OK
18. Press OK
19. Wait for the navmesh to generate
20. Press the Save Navmesh button
21. Save it into your project's NavMesh folder with the name terrainName_lowRes
22. Go back to the Navmesh tool
23. Next to Navmesh, select BTRlike from the dropdown
24. Press Generate
25. Select BTRlike from the drop down if it is not selected already
26. Press OK
27. Press OK
28. Wait for the navmesh to generate
29. Press the Save Navmesh button
30. Save it into your project's NavMesh folder with the name terrainName_btr
31. Go to your SCR_AIWorld entity
32. Find the first NavmeshWorldComponent
33. Under Navmesh Settings, find Navmesh Files Config and select set class
34. Press the .. button next to the Navmesh File text field
35. Select your Soldiers Navmesh
36. Find the second NavmeshWorldComponent
37. Under Navmesh Settings, find Navmesh Files Config and select set class
38. Press the .. button next to the Navmesh File text field

39. Select your BTR Navmesh
40. Find the third NavmeshWorldComponent
41. Under Navmesh Settings, find Navmesh Files Config and select set class
42. Press the .. button next to the Navmesh File text field
43. Select your lowRes Navmesh

Conflict (WIP)

This section is currently copy and pasted directly from a message from ThatRedGuy on the #reforger_conflict Discord channel. I have not tried to implement it myself but wanted to put it here because I thought it might be helpful to others and will give me a starting point when I get to this section:

For anyone that wants to create their own conflict mode it is super easy to do, just go to the map you want it on and create a SubScene layer and save the world, then go to Plugins - Game Mode Setup. If the Template is not Conflict.conf click the two ".." and choose the Conflict.conf then click Next, Then click Scan World, after it is done click next, and then click Create Entities, then Next again and Create Header then click Yes to All then finish the menu and you will have almost complete conflict mode!

This will give you all the managers needed for the mode, all that is needed now is to place Bases and FIA

Bases will be - ArmaReforger - Prefabs - Systems - MilitaryBase - MilitaryBase.et (Drag and drop these onto your map to get your bases)

FIA will be in ArmaReforger - Prefabs - Systems - AmbientPatrols - AmbientPatrolSpawnpoint_FIA.et

Customize Base details in the SCR_CampaignMilitaryBaseComponent things like if the point is Control Point/HQ/Small base As well as the name as size of base

Change total Supplies in the SCR_CampaignSuppliesComponent and build range in the CampaignBuildingProviderComponent

FIA can be customized in the SCR_AmbientPatrolSpawnPointComponent where you can change group type or set to random and give a respawn timer if you want

Oh don't forget about radios and supply depots

Radio is ArmaReforger - Prefabs - Systems - MilitaryBase - ConflictRelayRadio.et

Supply Depot is ArmaReforger - Prefabs - MP - Campaign - CampaignRemnantsSpawnPointEntity.et

Import Shapes with Forest Generators

This section explains how to import polygons so that they get a forest generator applied to them on import. It is mostly useful for importing shapefiles from Arma 3 terrains or vector data of real world locations.

1. Save your work
2. In order for this method to work, each shape in your shapefile requires an attribute that defines which type of forest it is. The only setup I have tested so far is the attribute of ID with each value being a number. It's possible other values will work but this tutorial is going to use that setup. I recommend using QGIS to add the ID attribute and a value.
3. Go to your project directory and create a folder called forests inside of it
4. Save or copy your .shp (along with .shx, .prj, and .dbf) file into the yourProject\forests folder
5. Go to the Import Shapefile button in the top toolbar
6. The SHP Path field is a path to your shapefile. Press the ".." button.
7. In the Choose m_sHPPPath (shp files) window, select your shapefile
8. Press OK
9. If your shapefile was made for Arma 3 or from GTT, you will want to add an X offset value of -200000.
10. If your shapefile is from a different source, for example OSM data you downloaded, you will need to find the offset to use. If for example your shapefile is a square that covers your whole terrain and the coordinate of the bottom left vertex is 251,267 then your offset would be X: -251 and Z: -267.
11. In the ID Column Name field, type ID
12. Go to Resource Browser and search for "SCR_SHP". There should be one file called `SCR_SHPPrefabDataList_ForestGenerator.conf`
13. Right-click on the file and go to Duplicate to <yourProjectName>
14. Select a name for this file. I normally do SCR_SHP_ForestGenerator
15. Press OK
16. The file should now appear in your project folder inside of a folder called `Configs\ForestGenerator`
17. In Resource Browser, right-click on your new .conf file (ex. `SCR_SHP_ForestGenerator.conf`)
18. Go to Open in Resource Manager. This will show the contents of the file.
19. You can left-click on the blue > signs and it will open each class. Each class has an ID and a Prefab. Any shapefile that has an attribute ID of 1 will get the `FG_Forest_Spruce1.et` prefab assigned to it. An ID of 2 will assign `FG_Forest_Pine1.et`.
20. These IDs and Prefab are editable.
21. If you wish, you can add a new entry by going to the top line where it says (17) and clicking the + button.
22. A new entry will be added to the bottom. You can open it and assign an ID and prefab.
23. If you made changes, go up to Workbench and select Save
24. Go back to the World Editor

25. From the resource browser, drag your .conf file into the Import Shapefile tool and into the Prefab Data List field where it says “set class”
26. Select Import SHP and your shapes will be imported with forest prefabs assigned to them
27. If your forests do not appear over your terrain, you may need to use the offset fields to align them.
28. Save the work

Debug Menu

There is a menu with various different debugging options that can do things like hide water. Use Left Windows + Left ALT to open it and then use the arrow keys and backspace to navigate through it. Hiding water can be found under Render > Ocean > Ocean Render

Export Shapes

You can export geographic data from World Editor.

1. Select the Export Geographic Data option in the Toolbar
2. Press the “..” next to Export Path
3. Point to any file in your project directory
4. Rename the filename in the Export Path field
5. Select which file type you want to export to; GeoJSON or SVG
6. In the Hierarchy window, select whichever items you want to export
 - a. Note: You can’t select Layers. You must select all the items in the layer.
7. Press Export
8. To load in QGIS, set the project CRS to EPSG:32631 (lower right of window)
9. Load the geojson
10. Right-click on the geojson in Layers
11. Go to Properties
12. Go to Source
13. Change the CRS to EPSG 32631

Appendix A: New Engine Features

- Engine now supports above ground water
- Support for non-power of 2 heightmap grids
- Support for non-square heightmaps

Appendix B: Tips and Tricks

Surface Painting

When painting surfaces, using Ctrl + X is a useful toggle to show the detailed textures, or to switch to a color map view where each surface is assigned a color to easier show different, similarly colored textures.

Satellite Image

- To see the satellite image when up close to the terrain:
 - Select your Terrain entity
 - Go to Object Properties
 - Turn “Middle Distance Max” all the way down

Add multiple start locations to Game Master

It's possible to add multiple start locations to Game Master like Everon and Arland have.

1. Make sure you have Game Master setup correctly
2. Go to the GameMode_Editor_Full entity in your Game Master subscene
3. Right-click on SCR_GameModeEditor
4. Select Add Component
5. Search for SCR_EditorCameraDefaultsComponent
6. Select SCR_EditorCameraDefaultsComponent
7. Press the + button next to Positions to add a new position
8. Move your camera to the position and angle you want to use for your start location
9. Look down at the bottom of the World Editor window and you should see a set of three coordinates and three directions
10. Use the three coordinates to fill out the X, Y, and Z coordinates for your Camera Position
11. Use the three directions to fill out the X, Y, and Z angles for your Camera Angles
12. You should see a pink pyramid like shape appear at the camera position
13. Repeat steps 7 through 11 for as many positions as you want

Snapping objects

1. Hold down B to view the vertices and drag one vertex towards another object
2. The object should snap to the other object's vertex

Creating copies of objects

1. Select the Move Tool (W)
2. Select the object you want to make a copy of in the game world
3. Hold down Alt and drag one of the arrows
4. A copy should be created

Poles on vertices for Powerlines

- If you want poles for the powerline to be placed on the vertices of your polyline, when you create your initial shape, set the first two vertices to Generate Per Point and each vertex added will have a powerline on it.

Appendix C: Troubleshooting

Terrain is not at 0, 0, 0 and objects are already placed

1. Save your work.
2. Put all your objects as a child of your Terrain Entity.
3. Set the Coords of your terrain entity to 0, 0 ,0
4. Your objects should follow

Cannot hear any environmental sounds

Make sure you have a soundmap generated and an AmbientSounds prefab

Soldiers will not move or not move far

Make sure you have a soldier and lowres navmesh set in your AIWorld

Can't move the camera past a certain point

Try restarting World Editor and see if that works.

Terrain has a big square over part of it

Regenerate the Normal Map by going to the Terrain Tool, pressing Generate normal map, and pressing OK.

Vehicle can't move when previewing

Make sure your Terrain Entity has the **Terrain** checkbox checked by Selecting your Terrain entity from the Hierarchy panel, going to the Object Properties Window and verifying that **Terrain**, under **Layers** is checked.

Satellite image is pink

Make sure your image format is .png and not something like .jpg. This might be the case if you downloaded your image off of <https://heightmap.skydark.pl/>

When the weather is set to Cloudy, the sky is black

There are two problems that can cause this:

1. In the Environment section of the world entity, if the planets are not in the order of Sun, Moon, Stars.
2. The GMT time zone not matching the coordinates.

After updating to 1.0, saved Object Brush no longer works

The format of the object brush has changed between pre-1.0 versions and version 1.0. The Old format used `ObjectBrushObjectBase` for each object in the brush but after 1.0 the new class is `SCR_ObjectBrushObjectBase`. You can manually open your Object Brush in a text editor and add the `SCR_` before each `ObjectBrushObjectBase` class to make it `SCR_ObjectBrushObjectBase`.

My Object Brush .conf is empty when opened in Workbench.

Try creating and saving a new brush and then comparing the old one that isn't working in a text editor to the new one. This is how the missing `SCR_` was discovered when 1.0 was released.

Manually modifying your old brush to match the new one will generally work but make sure you back up your original.

Grass is matching the color of the satellite map.

This is by design and having weird colors in your satellite image can cause weird colors to show up on clutter or surfaces. The best thing to do is to remove these unnatural colors from your satellite imagery. If that is not what you want to do you can go through the following steps to remove the satellite image blending from your surfaces.

1. Back up your project!!!!
2. Find the material you are using in Resource Browser (Terrains > Common > Surfaces)
3. Right-click on it and select Inherit in <Terrain Name>
4. Rename it (I just put a "b" after it)
5. Locate the inheritance you just made
6. Double-click it and go to Sat Map Blend and Sat Map Brightness and set them to 0
7. Find Plant Mat by scrolling up (or down from the Sat options)
8. Find that config in Resource Browser and inherit from that one again
9. Rename it again
10. Set it in the Plant Map field
11. Hit the little box with the arrow (or find it in resource browser and open it that way)
12. Find Sat Map Lerp and Sat Map Brightness and set them to 0
13. Save this file
14. Save the Surface material you made too
15. Go to Terrain Tool
16. Go to Paint
17. Got to your grass surface
18. Right-click on it and go to Change Layer Material
19. Select your material

Objects Don't Load

Check and see if you have a particle emitter present on your terrain. The log may show this.

Main Viewport is dark even on default terrains

1. Check your console logs for messages about "Invalid shader"
2. If that message appears, check and see if you are using Arma Reforger Experimental Tools but loading from the base game

Blocks of ground have weird color

1. Select your terrain tool

2. Select the “Info & Diags” tab
3. Select the terrain block that has the color issue. (Make sure it has a red outline to confirm you selected it). There may be a paint that has 0.0% or another relatively low percentage.
4. Right click the low percentage paint and merge it with whatever paint you’d like to merge it with (preferably your grass if so).
5. This should fix the issue, if it does not, merge it all into one paint or remove all paints and repaint the surface.
6. Save your world

Model underground is fully or partially lit despite the presence of a light probe

Make sure the light probe is marked as priority

Appendix D: Everon

- How to open in World Editor
 - a. Launch Arma Reforger Tools
 - b. Create New
 - c. Give it a Project Name of Everon_Preview
 - d. Press Ok
 - e. Go to Editors > World Editor
 - f. In the Resource Browser, navigate to ArmaReforger > worlds > Eden
 - g. Double-click Eden.ent
- Information
 - a. 2m heightmap
 - b. 1m imagery/surfaces
 - c. 6400x6400px heightmap grid
 - d. 12.8km x 12.8km

Appendix E: QGIS (WIP)

Temporary Placeholder for OSM Buildings > World Editor. Do a search for “You might need to adjust some buildings btw. OSM isn't always great” in enfusion_terrain_makers